

NET-002 — Cisco Inter-VLAN Routing Lab

Network Lab Documentation

Project Type: Networking Lab / Infrastructure Configuration/ Routing

Platform: Cisco Packet Tracer

Technician: Ridley Lherisse

Date Started: 10/06/2026

Date Completed: 10/06/2026

1. Project Overview

Objective

Describe what the lab was designed to accomplish.

Example:

The objective of this project was to configure VLAN segmentation and enable communication between separate network segments using inter-VLAN routing. This lab demonstrates Layer 2 switching, 802.1Q trunking, and Layer 3 routing concepts using Cisco Packet Tracer.

2. Environment & Tools

Software:

- Cisco Packet Tracer

Network Devices:

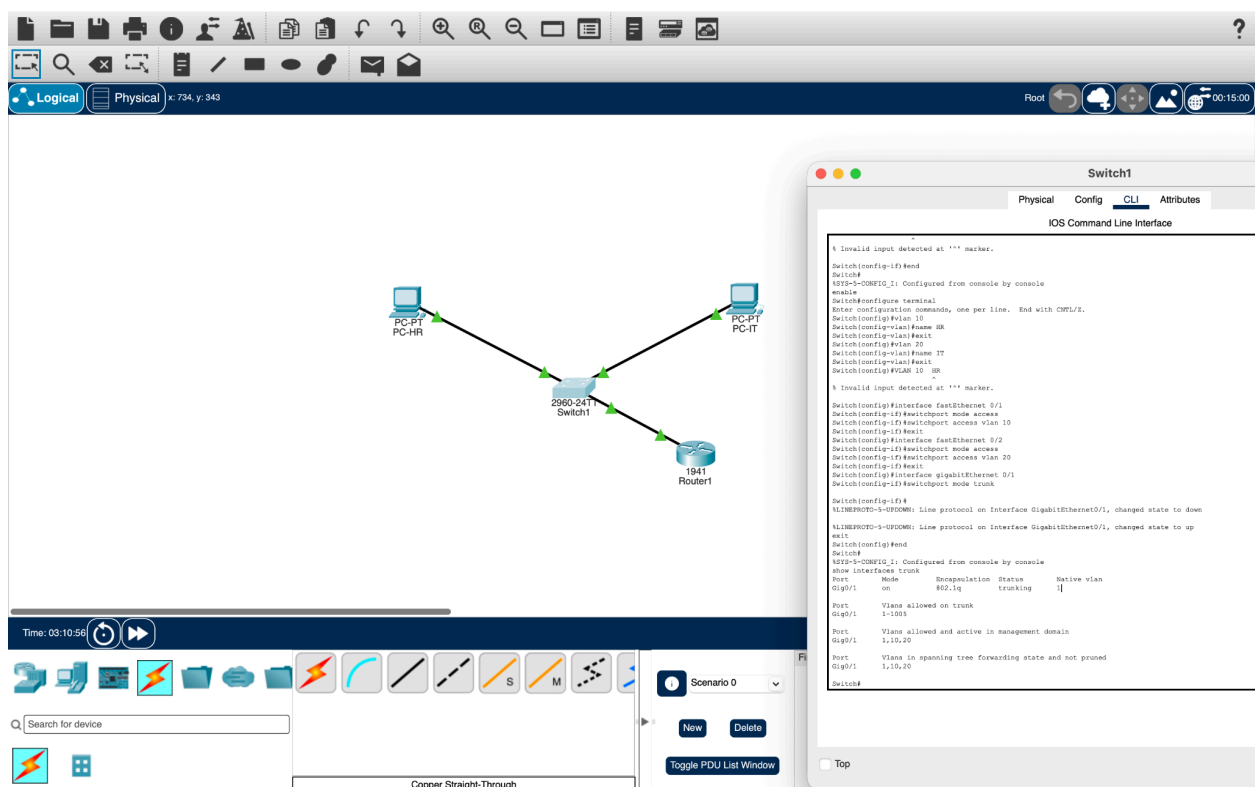
- Cisco 2960 Switch
- Cisco 1941 Router

Endpoints:

- PC-HR
- PC-IT

3. Network Design

Topology



Example:

PC-HR

|

VLAN 10

|

Switch

|

802.1Q Trunk

|

Router
|
VLAN 20
|
PC-IT

4. Network Configuration

VLAN Configuration

VLAN	Name	Purpose
10	HR	Department Network
20	IT	Department Network

Switch Configuration

Configured:

- Access ports assigned to VLANs
- 802.1Q trunk link enabled
- VLAN traffic allowed between switch and router

Example:

```
interface GigabitEthernet0/1
switchport mode trunk
```

Router Configuration

Configured router on a stick using subinterfaces:

VLAN 10 Gateway

```
interface GigabitEthernet0/0.10
encapsulation dot1Q 10
ip address 192.168.10.1 255.255.255.0
```

VLAN 20 Gateway

```
interface GigabitEthernet0/0.20
encapsulation dot1Q 20
ip address 192.168.20.1 255.255.255.0
```

5. Troubleshooting Process

Issue

Example:

Initial communication between VLAN 10 and VLAN 20 was unsuccessful.

Investigation

Steps performed:

- Verified physical connectivity
- Checked switch VLAN assignments
- Verified trunk configuration
- Checked router interface status
- Corrected interface configuration issues

Resolution

Enabled proper trunk communication and configured router subinterfaces to provide Layer 3 routing between VLAN networks.

6. Testing & Verification

Test 1 — Trunk Verification

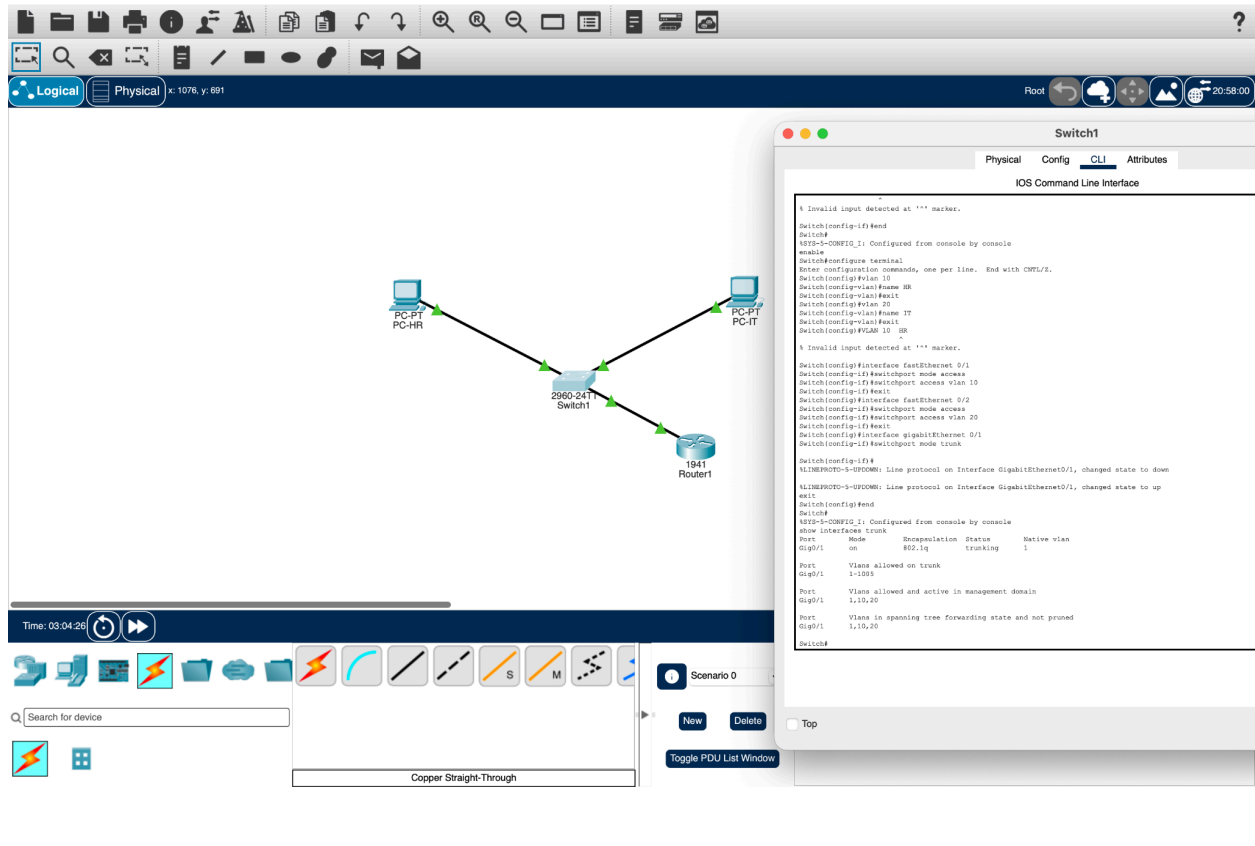
Command:

```
show interfaces trunk
```

Result:

Gig0/1 trunking
802.1Q encapsulation

Screenshot:



Test 2 — Interface Verification

Command:

show ip interface brief

Result:

GigabitEthernet0/0.10
192.168.10.1

GigabitEthernet0/0.20
192.168.20.1

Test 3 — Inter-VLAN Connectivity

Test: PC-HR ping PC-IT

Result: Successful replies received

7. Skills Demonstrated

- VLAN Configuration
 - Cisco Switching
 - 802.1Q Trunking
 - Router-on-a-Stick Configuration
 - Layer 3 Routing
 - Network Troubleshooting
 - Command Line Interface (CLI)
 - Network Documentation
-

8. Lessons Learned

Example:

This project strengthened my understanding of how enterprise networks separate departments using VLANs. Troubleshooting the physical and logical layers reinforced the importance of validating connectivity from Layer 1 through Layer 3.

9. Final Status

Status: Completed 

Documentation Type: Network Infrastructure Lab